

Positive Last-Layer Lifting in Quartic Zero Repair

Signed transport normal forms, common-prefix insurance, and the remaining anchor lemma

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Abstract

We isolate a concrete positivity obstruction in a quartic zero-repair problem. The zero-repair recursion is a history-indexed transport constraint: a block endpoint coupling may be feasible even when no compatible collection of one-step conditional couplings has yet been supplied. This distinction is closely related in spirit to causal and bicausal optimal transport, where the information structure carried by a filtration is part of the admissibility constraint [5, 6, 4, 8].

For the explicit pair

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right),$$

we have

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

The main analytic result of this note is a signed last-layer normal form: all omitted prefix residuals can be pushed to the final independent prefix layer without increasing their history-weighted one-dimensional Kantorovich–Rubinstein cost. The proof is solver-free. It shows that the remaining obstruction is not transport itself but positivity: the signed push need not preserve nonnegative coefficients.

We then prove a common-prefix reserve lemma. A single independent reserve, paid before the relevant history is known and convolved through the already exact prefix tree, preserves every restored prefix equality. At the last layer, a one-dimensional stochastic-dominance condition is sufficient to construct positive viable children by monotone coupling. For the quartic pair there is an explicit universal Bernoulli reserve

$$R_{1/3} = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1$$

for which

$$Q * R_{1/3} \succeq_{\text{st}} P.$$

Thus the prefix problem is reduced to one sharply stated open question: whether one can choose a positive last-layer lift and a common viable anchor at every scale. Finite LP audits at $n = 3, 4$ support this possibility, but they are reported only as numerical evidence. No unconditional logarithmic upper bound is claimed.

1 Introduction: the principle before the machinery

1.1 Why a prefix problem appears

Ordinary optimal transport compares probability laws through couplings and transport costs; the Kantorovich–Rubinstein duality is the standard one-dimensional tool behind the W_1 estimates used below [1]. In a multistage problem, however, matching the final law is not

the same as respecting the information revealed along the way. This distinction is central in nested distance and causal/adapted transport [4, 5, 6, 7, 8]. The present zero-repair model is a specialized finite-alphabet recursion of that general type: every history must admit its own compatible continuation.

The surrounding quartic programme and the block/prefix decomposition were developed in the companion notes [13, 14]; the broader causal-information programme is described in [12]. The purpose of the present paper is narrower. It identifies exactly what remains difficult in the *prefix* part.

A block relaxation remembers only the endpoint identity of a whole first block. It therefore permits intermediate one-step equalities to fail. Write that failure at history h as

$$r_h = Q * B_h - \sum_x P(x) \delta_x * B_{hx}.$$

The prefix problem asks how much additional root inventory is required to restore

$$r_h = 0$$

at every internal history.

1.2 The mechanism in four moves

The entire note can be read as the following four-step reduction.

1. Signed transport is easy. If negative intermediate coefficients are temporarily allowed, a residual r_h can be pushed one generation downward. Convolution by the probability law Q is nonexpansive in W_1/KR , so the history-weighted transport cost does not increase.

2. Therefore all old prefix debt can be moved to one layer. Iterating the push gives a signed normal form

$$r_h = 0 \quad (1 \leq |h| \leq n-2),$$

with all remaining defect concentrated on $|h| = n-1$. Thus the high-dimensional “many residuals at many depths” picture is not the true obstruction.

3. Positivity is the wall. The signed correction can make a child law negative. Hence the real theorem must show that a comparable normal form exists inside the positive probability cone. The finite LP decider tests exactly this statement; it does not replace its missing proof.

4. Positivity can be insured before the history is known. Convolve the already exact prefix tree with one common reserve R . Because convolution commutes with the zero-repair recursion, every old equality remains exact. At the last layer it is enough to dominate one common viable anchor. For the quartic pair,

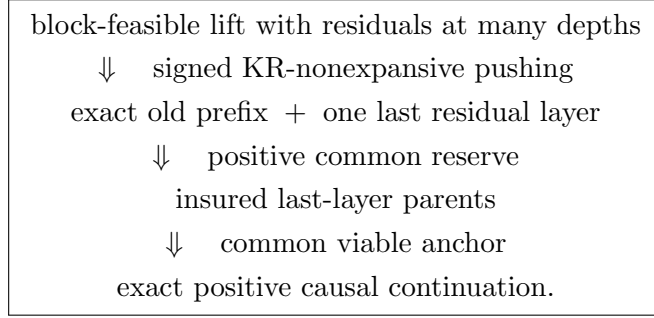
$$R_{1/3} = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1$$

satisfies the exact algebraic inequality

$$Q * R_{1/3} \succeq_{\text{st}} P.$$

The remaining problem is therefore a positive common-anchor selection problem, not a search for another abstract norm or spectral contraction.

Schematically,



This diagram is the conceptual content of the paper. The remaining sections make each arrow precise and identify which arrow is proved, numerical, or open.

1.3 What is proved, what is numerical, and what is open

Statement	Status
Signed residuals can be pushed to the last independent prefix layer without increasing weighted KR cost	PROVED
The obstruction left by that push is coefficientwise positivity (and, in finite truncations, support compatibility)	PROVED
Positive last-layer normal forms exist in the audited finite LPs $n = 3$ and $n = 4$ at the stated tolerances	NUMERICAL
A common reserve preserves all previously restored prefix equalities	PROVED
The explicit Bernoulli reserve $R_{1/3}$ satisfies $Q * R_{1/3} \succeq_{\text{st}} P$	PROVED
A positive common viable anchor exists uniformly at every scale	OPEN
The prefix premium is uniformly bounded by $1/3$	OPEN
The full global $O(\log n)$ upper bound	OPEN

This status separation is important. Computation is used here to falsify or support finite claims, while asymptotic statements are accepted only when an analytic argument is supplied.

2 Formal setup and logical status

Let

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right)$$

on the alphabet

$$\mathcal{A} = \{0, 1, 2, 3, 4\}.$$

Their generating functions satisfy

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}. \tag{1}$$

In particular, P and Q have the same mean and their PGF jets agree through order three at $z = 1$.

The quartic cancellation is specific to the present construction; it is not a generic consequence of causal transport theory. The general transport literature supplies the surrounding language—couplings, stochastic order, W_1 duality, and information-respecting multistage transport—whereas the normal-form and insurance identities below are specific statements proved for this zero-repair recursion.

A depth- N zero-repair tree is a family of probability laws

$$\{B_h\}_{|h| \leq N}$$

on $\mathbb{Z}_{\geq 0}$, indexed by source histories $h = (x_1, \dots, x_t) \in \mathcal{A}^t$, satisfying

$$\boxed{Q * B_h = \sum_{x \in \mathcal{A}} P(x) \delta_x * B_{hx}} \quad (|h| < N). \quad (\text{ZR})$$

The finite-horizon zero-repair bill is

$$\mathcal{B}_N := \inf \left\{ \mathbb{E}_{B_\emptyset} J : \{B_h\}_{|h| \leq N} \text{ satisfies (ZR)} \right\}.$$

The global upper-bound programme separates a block-endpoint term from a genuinely causal prefix-refinement premium. The present note addresses only the latter. The statements proved below do *not* by themselves establish a matching global $O(\log N)$ upper bound.

Status convention. Analytic lemmas are stated as theorems or propositions. Finite LP calculations are explicitly labelled numerical. The uniform positive last-layer/common-anchor statement remains open.

3 Block relaxation and prefix residuals

Fix a first-block length n . The block relaxation keeps the n -step endpoint identity but removes the internal one-step prefix equalities inside the first block. The exact block/prefix decomposition, finite-LP dual gap, and shadow-work representation motivating this relaxation are recorded in [13]; the related transport-tail and repaired-circulation viewpoint is developed in [14].

For a lifted block-feasible family define, at every omitted prefix node, the residual

$$\boxed{r_h := Q * B_h - \sum_x P(x) \delta_x * B_{hx}.} \quad (2)$$

Since all B_h are normalized,

$$r_h(\mathbb{Z}_{\geq 0}) = 0.$$

For a history

$$h = (x_1, \dots, x_t),$$

write

$$p(h) := \prod_{i=1}^t P(x_i), \quad p(\emptyset) := 1.$$

For a zero-mass signed measure ν of finite first moment on $\mathbb{Z}_{\geq 0}$, let

$$\|\nu\|_{\text{KR}}$$

denote its one-dimensional Kantorovich–Rubinstein norm; see, e.g., [1]. In one dimension the W_1 distance has the familiar cumulative-distribution representation used here. Equivalently, if

$$C\nu(j) := \sum_{i \leq j} \nu(i),$$

then

$$\|\nu\|_{\text{KR}} = \sum_{j \geq 0} |C\nu(j)|. \quad (3)$$

The independent prefix filling cost of a lifted block solution is

$$\mathfrak{F}_n(B) := \sum_{1 \leq |h| < n} p(h) \|r_h\|_{\text{KR}}. \quad (4)$$

The root residual is omitted: in the common lifted formulation it is algebraically redundant once the block endpoint identity and the independent prefix rows are imposed.

4 A signed last-layer normal form

We first remove positivity and work in the vector space of normalized signed measures. This exposes the exact algebraic mechanism.

Lemma 1 (Convolution is KR-nonexpansive). *Let Q be a probability law on $\mathbb{Z}_{\geq 0}$. For every zero-mass signed measure ν of finite first moment,*

$$\|Q * \nu\|_{\text{KR}} \leq \|\nu\|_{\text{KR}}.$$

Proof. By Kantorovich duality,

$$\|Q * \nu\|_{\text{KR}} = \sup_{\text{Lip}(f) \leq 1} \langle f, Q * \nu \rangle = \sup_{\text{Lip}(f) \leq 1} \langle Q^* f, \nu \rangle.$$

If f is 1-Lipschitz, then

$$(Q^* f)(j) = \sum_y Q(y) f(j + y)$$

is again 1-Lipschitz. Hence the final supremum is bounded by $\|\nu\|_{\text{KR}}$. $\square$

Lemma 2 (One-step signed residual push). *Assume $P(0) = p_0 > 0$. Let h be an omitted prefix node and let r_h be defined by (2). In the signed relaxation define*

$$B'_{h0} = B_{h0} + \frac{1}{p_0} r_h, \quad B'_{hx} = B_{hx} \quad (x \neq 0).$$

Then B'_{h0} is normalized,

$$r'_h = 0,$$

and the only next-layer residual changed directly by this operation is

$$r'_{h0} = r_{h0} + \frac{1}{p_0} Q * r_h.$$

Moreover,

$$p(h0) \left\| \frac{1}{p_0} Q * r_h \right\|_{\text{KR}} \leq p(h) \|r_h\|_{\text{KR}}. \quad (5)$$

Proof. The residual has zero total mass, so normalization is preserved. Also,

$$\begin{aligned} r'_h &= Q * B_h - \sum_x P(x) \delta_x * B'_{hx} \\ &= r_h - p_0 \frac{1}{p_0} r_h = 0. \end{aligned}$$

Equation (2) follows by substituting the modified law into the definition of the residual of $h0$. Finally,

$$p(h0) = p(h)p_0,$$

and Lemma 1 gives

$$p(h0) \left\| \frac{1}{p_0} Q * r_h \right\|_{\text{KR}} = p(h) \|Q * r_h\|_{\text{KR}} \leq p(h) \|r_h\|_{\text{KR}}.$$

□

Theorem 3 (Signed last-layer normal form). *Let*

$$\{B_h\}_{|h| \leq n}$$

be a normalized signed block lift with finite first moments, and suppose $P(0) > 0$. Then there exists another normalized signed lift

$$\{\tilde{B}_h\}_{|h| \leq n}$$

with the same root law and the same depth- n boundary laws such that

$$\boxed{\tilde{r}_h = 0 \quad (1 \leq |h| \leq n-2),} \tag{6}$$

while

$$\boxed{\sum_{|h|=n-1} p(h) \|\tilde{r}_h\|_{\text{KR}} \leq \sum_{1 \leq |h| < n} p(h) \|r_h\|_{\text{KR}}.} \tag{7}$$

Thus every signed prefix defect can be concentrated in the last independent prefix layer without increasing the total history-weighted KR cost.

Proof. Apply Lemma 2 to all residuals at depth 1, then to all residuals at depth 2, and continue successively through depth $n-2$. The history graph is a tree, so each residual is pushed only into descendants. At every stage the removed weighted KR contribution is replaced by a next-layer contribution of no larger weighted KR norm. The depth- n boundary laws are never modified. Iteration proves (6)–(7). □

Remark 4 (What the theorem does not preserve). Theorem 3 is an untruncated signed statement. It does not preserve coefficientwise nonnegativity, and it need not preserve a fixed finite support cutoff. These are precisely the two reasons why it is not by itself a theorem for the positive finite LP.

5 The positivity premium

Let $\mathcal{X}_{n,M}^{+, \text{blk}}$ denote a positive finite-support block-feasible face at cutoff M . On a fixed block-optimal face define

$$\mathfrak{F}_{n,M}^+ := \inf_{B \in \mathcal{X}_{n,M}^{+, \text{blk}}} \sum_{1 \leq |h| < n} p(h) \|r_h\|_{\text{KR}}.$$

Define the positive last-layer value

$$\mathfrak{F}_{n,M}^{+, \text{last}} := \inf_{\substack{B \in \mathcal{X}_{n,M}^{+, \text{blk}} \\ r_h=0, 1 \leq |h| \leq n-2}} \sum_{|h|=n-1} p(h) \|r_h\|_{\text{KR}}.$$

Then

$$\mathfrak{F}_{n,M}^+ \leq \mathfrak{F}_{n,M}^{+, \text{last}}.$$

Definition 5 (Positivity premium).

$$\Gamma_{n,M} := \mathfrak{F}_{n,M}^{+, \text{last}} - \mathfrak{F}_{n,M}^+ \geq 0.$$

The signed theorem says that the corresponding signed premium vanishes identically:

$$\Gamma_n^{\text{signed}} = 0.$$

Therefore any nonzero $\Gamma_{n,M}$ is generated entirely by the positive cone and/or the finite support interface.

This is the point at which the problem ceases to be a transport identity: one must carry the exact equalities

$$r_h = 0$$

through coefficientwise nonnegativity.

6 Finite positive decision experiments

This section contains computational evidence only.

The finite LP decider imposes the hard constraints

$$r_h = 0 \quad (1 \leq |h| \leq n-2)$$

and minimizes the KR defect only on the last independent prefix layer. Boundary laws are recursively audited for depth- n viability.

n	M	$\mathfrak{F}_{n,M}^+$	$\mathfrak{F}_{n,M}^{+, \text{last}}$	difference	boundary accepts
3	8	0.0697761219022406	0.0697761219025007	2.60×10^{-13}	0
3	10	0.0697761218851941	0.0697761218923195	7.13×10^{-12}	1
4	8	0.0739180923076735	0.0739182643909350	1.72×10^{-7}	5

For $n = 4, M = 8$, the constrained solution has

$$\begin{aligned} \text{depth 1 : } & 1.2353 \times 10^{-15}, \\ \text{depth 2 : } & 1.0918 \times 10^{-15}, \\ \text{depth 3 : } & 0.0739182643909363 \end{aligned}$$

in history-weighted KR defect. Thus the older layers satisfy the hard equalities to numerical precision.

The $n = 4, M = 8$ equality decision used tolerance 2×10^{-7} . The observed gap

$$1.72 \times 10^{-7}$$

is below that threshold but not negligible relative to it. Accordingly the correct statement is:

NUMERICAL: positive last-layer normal form verified at the stated finite-LP tolerance.

No exact rational equality

$$\Gamma_{4,8} = 0$$

is claimed.

7 A common-prefix reserve

The last-layer reduction suggests that positivity should be paid once, upstream, rather than repaired after individual histories are revealed.

We use the standard stochastic order on $\mathbb{Z}_{\geq 0}$ [2, 3]:

$$\mu \succeq_{\text{st}} \nu \iff F_\mu(t) \leq F_\nu(t) \text{ for all } t.$$

Let $\mathcal{K}_n$ denote the set of probability laws that admit a positive exact zero-repair continuation of depth n .

We use the following elementary property.

Lemma 6 (Stochastic upward closure). *If $C \in \mathcal{K}_n$ and $D \succeq_{\text{st}} C$, then $D \in \mathcal{K}_n$.*

Proof sketch. By Strassen's coupling theorem [2], couple $J_D \sim D$ and $J_C \sim C$ with $J_D \geq J_C$ almost surely. Carry the nonnegative excess inventory through an exact continuation of C . Since every zero-repair identity is translation-covariant in the buffer coordinate, the excess can be retained without destroying any equality. Disintegration yields a valid continuation with root law D . $\square$

Theorem 7 (Common-prefix reserve lemma). *Assume a positive block lift satisfies*

$$r_h = 0 \quad (1 \leq |h| \leq n-2).$$

Let R be a probability law of finite mean.

Convolve every prefix law through depth $n-1$ with the same reserve:

$$\widehat{B}_h := B_h * R, \quad |h| \leq n-1.$$

Then all already exact prefix equalities remain exact.

Assume further that there exists a common anchor

$$C_n \in \mathcal{K}_n$$

such that for every last-layer history $|h| = n-1$,

$$\boxed{(Q * B_h) * R \succeq_{\text{st}} P * C_n.} \tag{8}$$

Then the first block can be completed to exact positive depth- n boundary children belonging to $\mathcal{K}_n$.

The root mean increases by exactly $\mathbb{E}R$.

Proof. For every old exact node,

$$Q * B_h = \sum_x P(x) \delta_x * B_{hx}.$$

Convolution by the common reserve gives

$$Q * (B_h * R) = \sum_x P(x) \delta_x * (B_{hx} * R),$$

so the equality survives unchanged.

Fix now $|h| = n-1$. Let

$$S \sim (Q * B_h) * R.$$

Independently sample

$$X \sim P, \quad J \sim C_n,$$

so that $X + J \sim P * C_n$. By (8) there is a coupling with

$$S \geq X + J \quad \text{a.s.}$$

Condition on $X = x$ and define

$$D_{hx} := \text{Law}(S - x \mid X = x).$$

Then

$$D_{hx} \succeq_{\text{st}} C_n,$$

hence $D_{hx} \in \mathcal{K}_n$ by Lemma 6. Moreover

$$\text{Law}(S) = \sum_x P(x) \delta_x * D_{hx},$$

which is the required exact ZR equality at the last prefix layer.

Finally,

$$\mathbb{E}(B_\emptyset * R) = \mathbb{E}B_\emptyset + \mathbb{E}R.$$

□

Corollary 8 (A sufficient stochastic-floor condition). *Under the hypotheses of Theorem 7, suppose*

$$C_n \in \mathcal{K}_n$$

and every last-layer parent satisfies

$$B_h \succeq_{\text{st}} C_n.$$

If a reserve R satisfies

$$Q * R \succeq_{\text{st}} P,$$

then the anchor condition (8) holds automatically.

Proof. By stochastic monotonicity of convolution,

$$Q * (B_h * R) = B_h * (Q * R) \succeq_{\text{st}} B_h * P \succeq_{\text{st}} C_n * P.$$

□

8 An exact one-third insurance law

The quartic pair admits an explicit reserve with mean $1/3$.

Theorem 9 (Exact one-third insurance identity). *Define*

$$R_{1/3} := \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1, \quad G_{R_{1/3}}(z) = \frac{2+z}{3}.$$

Then

$$\frac{G_P(z) - G_Q(z)G_{R_{1/3}}(z)}{1-z} = \frac{10 + 11z + 19z^2 + 17z^3 + 7z^4}{192}. \quad (9)$$

All coefficients on the right-hand side are strictly positive. Consequently

$$Q * R_{1/3} \succeq_{\text{st}} P. \quad (10)$$

The mean reserve cost is

$$\mathbb{E}R_{1/3} = \frac{1}{3}.$$

Proof. Direct polynomial multiplication gives

$$G_P(z) - G_Q(z) \frac{2+z}{3} = \frac{1-z}{192} (10 + 11z + 19z^2 + 17z^3 + 7z^4).$$

For a law μ on $\mathbb{Z}_{\geq 0}$,

$$\sum_{t \geq 0} F_\mu(t) z^t = \frac{G_\mu(z)}{1-z}.$$

Thus coefficientwise positivity of

$$\frac{G_P - G_{Q * R_{1/3}}}{1-z}$$

is exactly

$$F_P(t) - F_{Q * R_{1/3}}(t) \geq 0 \quad \forall t,$$

which is (10). □

Corollary 10 (Universal insured dominance). *For every probability law B ,*

$$\boxed{Q * (B * R_{1/3}) \succeq_{\text{st}} P * B.}$$

This identity is important conceptually: the constant $1/3$ is not inferred from the finite LP sequence. It is an exact algebraic insurance threshold available at every scale.

9 A recursive low-tail viability barrier

The viable anchor cannot be chosen arbitrarily. The following exact low-dimensional barrier illustrates the recursive geometry of $\mathcal{K}_n$.

Define

$$a_0 := 1, \quad a_{k+1} := \frac{8 + 16a_k}{27}.$$

Proposition 11 (Recursive two-coordinate viability barrier). *If $B \in \mathcal{K}_k$, then*

$$\boxed{B(0) + \frac{7}{27}B(1) \leq a_k.} \tag{11}$$

Moreover

$$\boxed{a_k = \frac{8}{11} + \frac{3}{11} \left(\frac{16}{27} \right)^k,} \tag{12}$$

so

$$a_k \downarrow \frac{8}{11}.$$

Proof. For $k = 0$ the assertion is immediate from normalization.

For the first nontrivial step, one-step viability implies

$$Q * B \succeq_{\text{st}} P.$$

At threshold $t = 1$,

$$F_{Q * B}(1) = \frac{27B(0) + 7B(1)}{64} \leq F_P(1) = \frac{3}{8}.$$

Hence

$$B(0) + \frac{7}{27}B(1) \leq \frac{8}{9} = a_1.$$

Now assume the claim for $\mathcal{K}_k$, and let $B \in \mathcal{K}_{k+1}$. Choose viable children $C_x \in \mathcal{K}_k$ satisfying the ZR identity. The coefficient of 0 gives

$$\frac{7}{64}B(0) = \frac{1}{8}C_0(0),$$

hence

$$C_0(0) = \frac{7}{8}B(0).$$

The coefficient of 1 gives

$$\frac{20B(0) + 7B(1)}{64} = \frac{1}{8}C_0(1) + \frac{1}{4}C_1(0).$$

Since

$$C_0(1) \leq 1 - C_0(0) = 1 - \frac{7}{8}B(0)$$

and

$$C_1(0) \leq a_k$$

by the induction hypothesis,

$$20B(0) + 7B(1) \leq 8 - 7B(0) + 16a_k.$$

Therefore

$$27B(0) + 7B(1) \leq 8 + 16a_k,$$

which is

$$B(0) + \frac{7}{27}B(1) \leq a_{k+1}.$$

Solving the affine recurrence yields (12). $\square$

Remark 12. Proposition 11 is only one facet family. It does not describe the full viability polytope $\mathcal{K}_n$, and no claim of facet completeness or sharpness for all n is made.

10 The remaining theorem

The signed normal form and the one-third reserve isolate the remaining prefix obstruction.

Open Problem 13 (Positive last-layer / one-third anchor lemma). *For every relevant block scale n , there exists a positive block-optimal, or uniformly $O(1)$ -near-optimal, lifted first block*

$$\{B_h\}_{|h| \leq n}$$

such that

$$r_h = 0 \quad (1 \leq |h| \leq n-2),$$

and there exists a common viable anchor

$$C_n \in \mathcal{K}_n$$

satisfying

$$\boxed{(Q * B_h) * R_{1/3} \succeq_{\text{st}} P * C_n \quad \text{for every } |h| = n-1.} \quad (13)$$

A stronger but simpler sufficient formulation is the existence of a common stochastic floor

$$C_n \in \mathcal{K}_n, \quad B_h \succeq_{\text{st}} C_n \quad \text{for every } |h| = n-1.$$

Corollary 8 would then imply (13) immediately.

Theorem 14 (Conditional prefix bound). *Assume Conjecture 13 at scale n , with no additional root cost beyond the block-optimal value. Then the prefix-refinement premium satisfies*

$$\boxed{\Pi_n \leq \frac{1}{3}.} \tag{14}$$

More generally, if the chosen positive last-layer form is within C_0 of the block optimum, then

$$\Pi_n \leq C_0 + \frac{1}{3}.$$

Proof. Convolve the exact prefix part with $R_{1/3}$. By Theorem 7, the old prefix equalities remain exact. The anchor hypothesis creates viable positive children at the last prefix layer. Each child then admits an exact continuation of depth n . The root mean rises by exactly

$$\mathbb{E}R_{1/3} = \frac{1}{3}.$$

□

11 Relation to a logarithmic global upper bound

The prefix theorem is only one branch of the global doubling estimate. Writing schematically

$$\mathcal{B}_{2n} - \mathcal{B}_n = \Delta_n^{\text{blk}} + \Pi_n,$$

the present note addresses Π_n .

If, in addition to a uniform form of Theorem 14, the separate block-endpoint term obeys

$$\Delta_n^{\text{blk}} \leq C_{\text{blk}}$$

uniformly on dyadic scales, then

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C_{\text{blk}} + \frac{1}{3}.$$

Dyadic iteration gives

$$\mathcal{B}_{2^k} \leq \mathcal{B}_1 + k \left(C_{\text{blk}} + \frac{1}{3} \right),$$

hence

$$\boxed{\mathcal{B}_n = O(\log n).}$$

Combined with a matching logarithmic lower bound from the companion lower-bound analysis, this would yield

$$\mathcal{B}_n = \Theta(\log n).$$

Remark 15 (No theorem inflation). The preceding paragraph is a conditional implication. The uniform positive anchor statement (13) is open, and the endpoint branch is analytically separate. Therefore this note does not claim an unconditional $O(\log n)$ upper bound.

12 Why the reduction is useful

The sequence of reductions changes the form of the problem.

Initially, the prefix obstruction is an exponentially large history-indexed family of omitted equalities. The signed last-layer theorem shows that, in the absence of positivity, all older residuals are algebraically disposable: they can be pushed to the last independent layer at no increase in the natural KR currency.

Thus the real obstruction is not the existence of a transport normal form. It is the intersection of that normal form with the positive probability cone.

The common-prefix reserve lemma then shows that positivity need not be repaired history by history after the history is known. A single reserve paid at the root can be propagated through the exact prefix subtree without breaking any old equality. All that remains is a simultaneous last-layer stochastic dominance problem.

Finally, the quartic pair supplies an exact universal reserve

$$R_{1/3} = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1.$$

The open prefix problem is therefore no longer “find a mysterious contraction”. It is the concrete existence question

$$\exists C_n \in \mathcal{K}_n \quad \forall |h| = n - 1 : \quad (Q * B_h) * R_{1/3} \succeq_{\text{st}} P * C_n.$$

This is a falsifiable positive lifting statement.

13 Proof ledger

Statement	Status
Quartic identity $G_P - G_Q = (1 - z)^4/64$	PROVED
KR contraction under convolution	PROVED
One-step signed residual push	PROVED
Signed last-layer normal form	PROVED
Positive last-layer form at $n = 3, M = 8, 10$	NUMERICAL
Positive last-layer form at $n = 4, M = 8$ at tolerance 2×10^{-7}	NUMERICAL
Common-prefix reserve lemma	PROVED
Exact Bernoulli 1/3 insurance identity	PROVED
Recursive low-tail barrier (11)	PROVED
Uniform positive last-layer/common-anchor lemma	OPEN
Uniform bound $\Pi_n \leq 1/3$	OPEN
Global $O(\log n)$ upper bound	OPEN

14 Reproducibility note

The finite positive normal-form calculations quoted above were obtained from deterministic sparse linear programs implemented in Python/NumPy/SciPy [9, 10]. The LP interface uses the HiGHS solvers; the dual revised simplex technology underlying HiGHS is described in [11]. The relevant decider imposes exact ZR rows on all prefix depths $1, \dots, n - 2$, minimizes KR defect only on depth $n - 1$, and recursively audits every boundary law for depth- n continuation.

The principal numerical outputs recorded in this note are:

- $n = 3, M = 8$:

$$\mathfrak{F}^+ = 0.0697761219022406, \quad \mathfrak{F}^{+, \text{last}} = 0.0697761219025007.$$

- $n = 3, M = 10$:

$$\mathfrak{F}^+ = 0.0697761218851941, \quad \mathfrak{F}^{+, \text{last}} = 0.0697761218923195.$$

- $n = 4, M = 8$:

$$\mathfrak{F}^+ = 0.0739180923076735, \quad \mathfrak{F}^{+, \text{last}} = 0.0739182643909350.$$

These values are evidence for the positive lifting conjecture. They are not substituted for the missing uniform analytic argument.

15 Conclusion

The prefix-refinement problem for the quartic pair admits a sharp structural reduction.

In the signed category, all prefix defects can be pushed to the last independent layer with no increase in history-weighted KR cost. Consequently the only obstruction to this normal form in the probability problem is positivity.

A common reserve can then be propagated through the exact prefix tree without breaking any old equality. The quartic algebra supplies an explicit scale-independent Bernoulli reserve of mean $1/3$, reducing the remaining positive problem to a common-anchor stochastic dominance condition.

The present frontier is therefore:

find, for every scale n , a positive last-layer normal form and a common $C_n \in \mathcal{K}_n$ such that $(Q * B_h) * R_{1/3} \succeq_{\text{st}} P * C_n \quad \text{for all } h = n - 1.$

Proving this statement would give a uniform constant bound on the prefix premium. The remaining passage to a global logarithmic upper bound would then require the separate block-endpoint estimate.

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